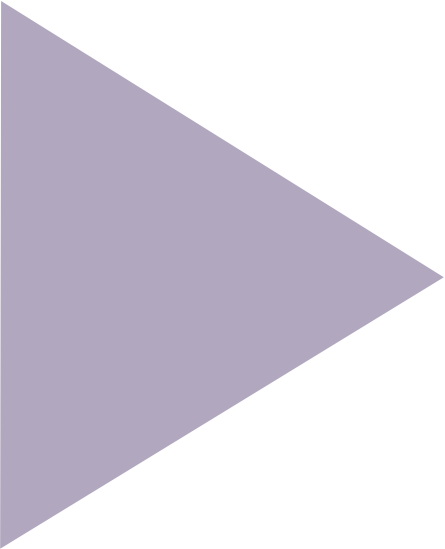




Tab 2: Value at Risk in Portfolio Management

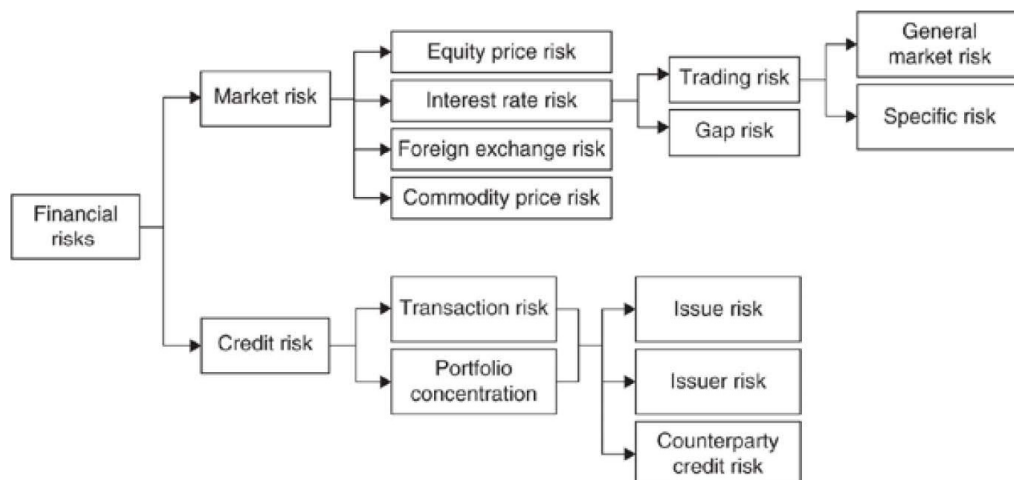
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Financial Risks



JP Morgan and 'Inventing VaR'

JPMorgan is credited with helping to make VaR a widely used measure. The Chairman, Sir Dennis Weatherstone was dissatisfied with the long risk reports he received every day. These contained a huge amount of detail on the Greek letters for different exposures, but very little that was really useful to top management. He asked for something simpler that focused on the bank's total exposure over the next 24 hours measured across the bank's entire trading portfolio.

At first, his subordinates said this was impossible, but eventually they adapted the Markowitz portfolio theory to develop a VaR report. This became known as the 4.15 report because it was placed on the chairman's desk at 4.15pm every day after the close of trading.

John C. Hull, *Risk Management and Financial Institutions*, Wiley, 2012

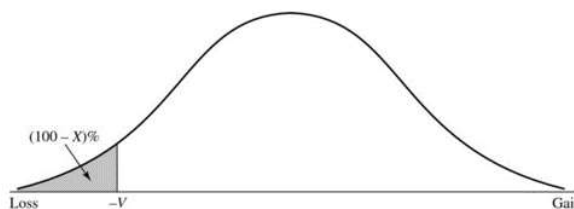
Value at Risk (VaR)

Definition: Value at Risk (VaR)

Value at risk (VaR) can be defined as the worst loss that might be expected from holding a security or portfolio over a given period of time (say one day, or 10 days for the purpose of regulatory capital reporting), given a specified level of probability (confidence level).

Example:

Let's say that a position has a daily VaR of \$10m at the 99% confidence level, we mean that the realised daily losses from the position will on average be higher than \$10m on only one day in every 100 trading days (i.e. two to three days each year).



Interpretation of VaR

VaR is not the answer to the question: How much can I lose on my portfolio over a given period of time? The answer to this question is 'everything', or any fraction value of the portfolio.

Instead, VaR offers a probability statement about the potential change in the value of a portfolio resulting from a change in market factors over a specified period of time.

Crucially, the VaR measure also does not state by how much actual losses are likely to exceed the VaR figure; it simply states how likely (or unlikely) it is that the VaR measure will be exceeded.

VaR and Time Horizons

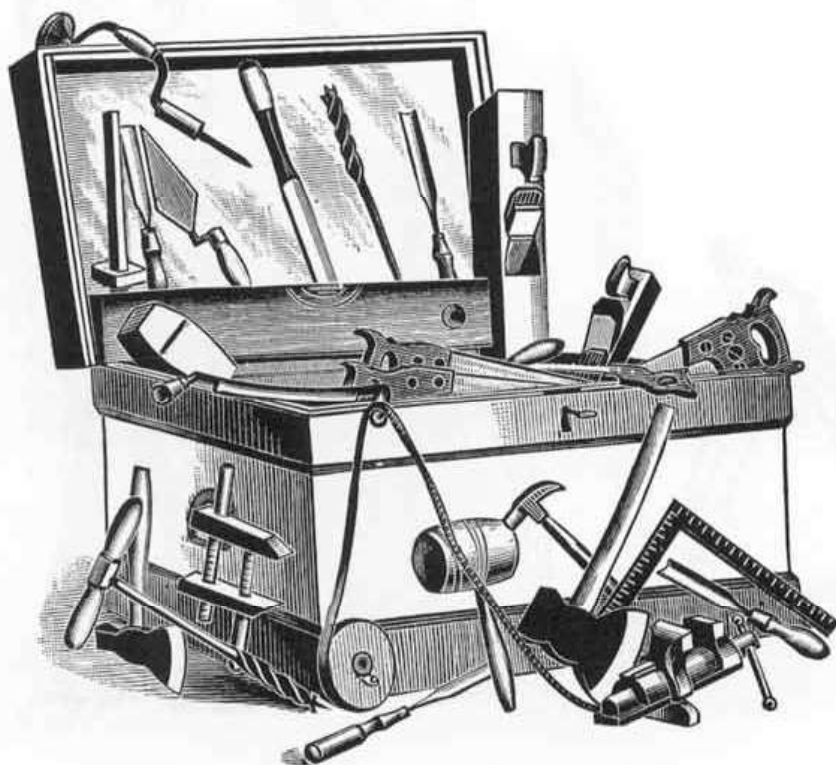
VaR is often used to manage market risk over a 1-day time horizon. For this purpose, it's necessary to derive VaR from the daily distribution of the portfolio values.

However, regulators have set a time horizon of 10 days for the purpose of VaR calculations that are used to report regulatory capital requirements. Ideally, this '10-day VaR' would be derived from a corresponding distribution of results over a 10-day horizon. This is problematic, however, as it implies that the time series of data used for the analysis must be much longer—indeed, 10 times longer—than that employed in any one-day VaR analysis.

As a result, many banks employ a work-around that allows them to derive an approximation of 10-day VaR from daily VaR data by multiplying the daily VaR by the square root of time (here, 10 days). The 'square root of time' rule is endorsed by the regulators.

$$T\text{-day VaR} = 1\text{-day VaR} \times \sqrt{T}$$

Methods for Estimating VaR



Methods for Estimating VaR

First method: Analytic variance-covariance approach (aka parametric VaR)

Under the analytic variance-covariance approach or 'Delta normal' approach, we assume that the risk factors and the portfolio values are log-normally distributed or, equivalently, that their log returns (the log of the returns) are normally distributed.

This makes the calculation much simpler, since the normal distribution is completely characterised by its first two moments, and the analyst can derive the mean and the variance of the portfolio return distribution from:

- a. The multivariate distribution of the risk factors
- b. The composition of the portfolio

$$\text{VaR} = \mu + \sigma N^{-1}(X)$$

Methods for Estimating VaR

Second method: Historical simulation (aka non-parametric VaR)

The historical simulation approach to VaR calculation is conceptually simple and does not oblige the user to make any assumptions about the distribution.

However, at least two or three years of historical data are necessary to produce meaningful results. Three steps are involved:

1. Select a sample of actual daily risk factor changes over a given period of time, say 500 days (i.e. two years' worth of trading days), using the same period of time for all the factors.
2. Apply those daily changes to the current value of the risk factors, revaluing the current portfolio as many times as the number of days in the historical sample. Sum these changes across all positions, keeping the days synchronised.
3. Construct the histogram of portfolio values and identify the VaR that isolates the first percentile of the distribution in the left-hand tail (assuming VaR is derived at the 99% confidence level).

Advantages

- The major attraction of historical simulation is that the method is completely nonparametric (i.e. we don't need to worry about setting parameters) and does not depend on any assumptions about the distribution of the risk factors.
- The nonparametric nature of historical simulation also obviates the need to estimate volatilities and correlations. Historical volatilities and correlations are already reflected in the data set, so all we need to calculate are the synchronous risk-factor returns over a given historical period.
- Historical simulation has also no problem accommodating fat tails in distributions, since the historical returns already reflect actual synchronous moves in the market across all risk factors.

Disadvantages

- The main drawback of historical simulation is its complete dependence on a particular set of historical data.
- The underlying assumption is that the past, as captured in this historical data set, is a reliable representation of the future.

Methods for Estimating VaR

Third method: Monte Carlo Simulation

Consists of repeatedly simulating the random processes that govern market prices and rates. Each simulation (scenario) generates a possible value for the portfolio at the target horizon (e.g. 10 days). It involves three steps:

- a. Specify all the relevant risk factors, their stochastic processes and parameter estimates.
- b. Construct price paths. Price paths are constructed using random numbers produced and advancing one step at a time (daily) the numerical solution to the stochastic processes.
- c. Value the portfolio for each path (scenario).

The process is repeated a large number of times, say 10,000 times, to generate the distribution, at the risk horizon, of the portfolio return. VaR at the 99% confidence level is then simply derived as the distance to the mean of the first percentile of the distribution, as for our other calculation methods.

Example:
VaR for One Asset



Example: VaR for One Asset

Position: USD100,000 in QQQ (Nasdaq 100)

Standard deviation of daily returns: 2.21%

99% confidence level: 2.33 standard deviations

Assignment:

1. Calculate 1-day VaR for this asset at 99% confidence
2. How would you then scale this up to 10 days?
3. List three limitations to the use of VaR as a measurement of market risk

VaR formula for a single asset = $\text{Vol} \times \text{Square Root [T]} \times \text{CL} \times \text{N}$

Vol: Daily volatility of the underlying asset

Square Root [T]: Square root of time (days)

CL: Confidence level (in standard deviations)

N: Notional or underlying amount

Example: VaR for One Asset

1-day VaR = $2.21\% \times \text{Square Root [1]} \times 2.33 \times \$100,000 = \$5,149$

10-day VaR = $2.21\% \times \text{Square Root [10]} \times 2.33 \times \$100,000 = \$16,284$

- Note: Complexities arise at portfolio level as we have to take into account correlations between different positions

Limitations:

- Past not always good indicator of the future
- May not cover unusual market conditions (e.g. LTCM, Lehman default)
- Some market risks are not easily captured by VaR e.g.
 - Non-linear pay-offs such as options
 - Managed rates
 - Relative value risks e.g. basis, correlation
- Assumptions about liquidity – time horizon may be inappropriate
- Assumes continuous, small moves in value (no discontinuity)

Market Risk: Basel III VaR Models Approach

- Base Value at Risk (VaR) Modelling Approach
- Stringent regulatory approval conditions
- Model parameters:
 - 99% confidence level
 - 10-day minimum holding period
 - Historical data covers at least 1 year
 - Multiplier of between 3 and 4 applied to model results
- Absolute capital \$/€/£ figure for market risk – converted into RWAs by multiplying by 12.5
- Back testing, stress and historical scenario testing (Pillar 2)
- From 2011 must include CVA, stressed VaR, incremental risk charge and liquidity horizons in capital charge

Expected Shortfall (ES)

- Expected value of all changes in the portfolio value in the tail of the distribution of gains and losses conditional on these changes exceeding the VaR
 - Requires a simulation (e.g. historical, Monte Carlo) to calculate
 - Also known as conditional VaR or expected tail loss
- Takes into account the severity of a loss rather than just its probability
- ES is a 'coherent' risk measure, VaR is not
 - Reflects the risk reduction from diversification effects ('subadditivity')
- ES just gives more detail about a particular distribution rather than all potential sources of risk
- ES is much more difficult to backtest than VaR



GE imagination at work

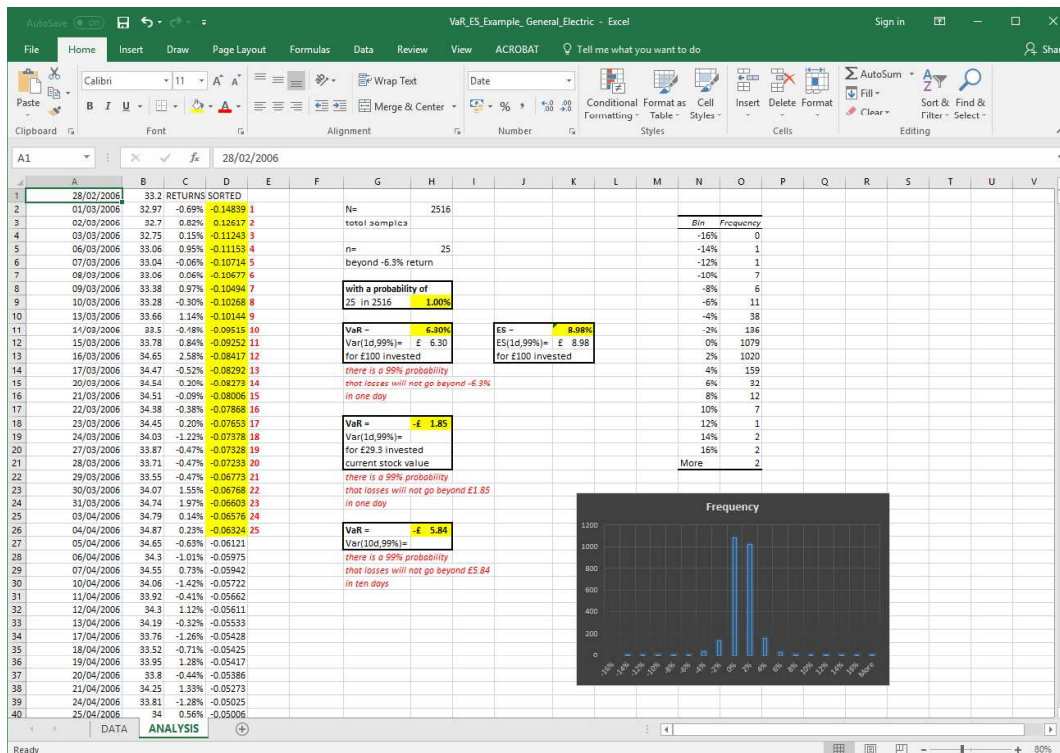
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Example:
The VaR and ES of
General Electric Corp.



Exercise



Backtesting Value at Risk

- Backtesting tests validity of VaR model
 - Regulatory requirement if using model to calculate capital
- Compare VaR figures to actual outcomes
- More rigorous approach is to compare VaR to 'hypothetical' P&L
 - Based on holding end of day position throughout next day

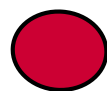
'Traffic lights' – Basel I Approach



< 5 exceptions pa



5-10 exceptions pa

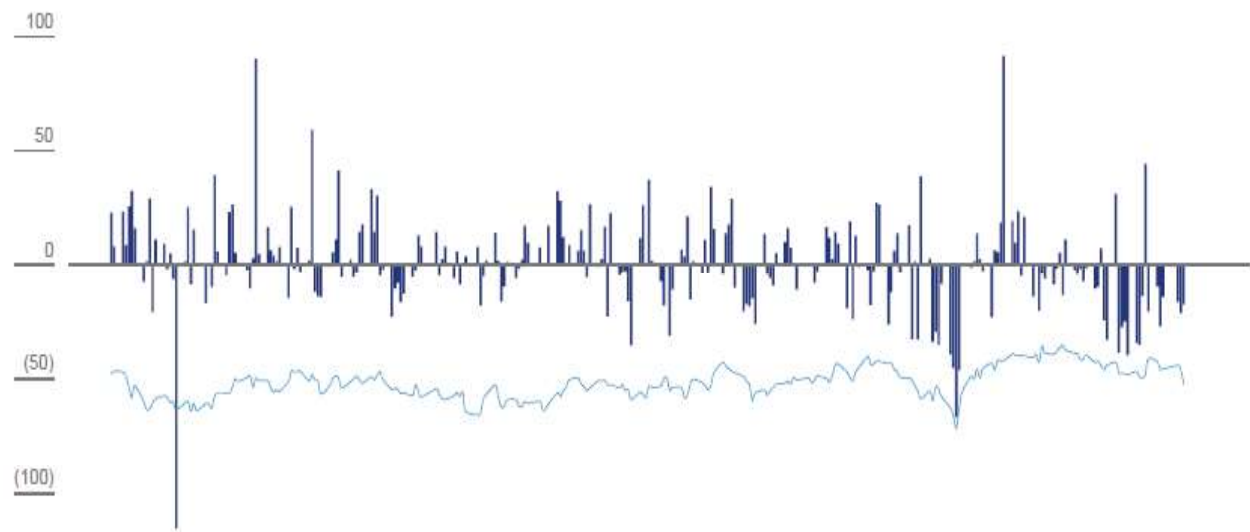


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Backtesting Value at Risk: Deutsche Bank

Comparison of trading units daily buy-and-hold income and value-at-risk in 2014

in € m.



Source: Deutsche Bank Annual Report 2014, page 194

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A Guide to IMF Stress Testing: Methods and Models

A Guide to IMF Stress Testing: Methods and Models
Ancillary Materials

To go back to the book, please click [here](#).

Toolkit Files

Note to readers: Ancillary materials are arranged based on the chapter in which they appear in the book.

The files listed below are also available on the companion CD.

Chapter 3
[Stress Tester 3.0](#)

Chapter 4
[Excel Spreadsheet Macro for the Breaking Point Method](#)

Chapter 5
[Excel Spreadsheet Macro for the Next-Generation Solvency Stress Test](#)

Chapter 6
[Excel Spreadsheet Macro for the Market and Funding Liquidity Stress Tests](#)

Chapter 7

Creazione di una connessione protetta in corso...

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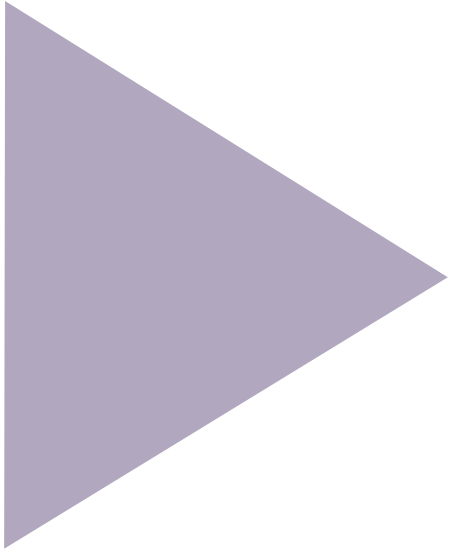
Case Study



'London Whale' Rattles Debt Market
by Wall Street Journal (Published on 6 Apr 2012)
https://www.youtube.com/watch?v=pen_3uYu0s4

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Decomposing Risk

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Risk Decomposition

Let $w = (w_1, w_2, \dots, w_N)'$ denote the vector of portfolio weights on N assets, instruments, asset classes, or managers, and let $\sigma(w)$ and $\text{VaR}(w)$ denote the portfolio standard deviation and value-at-risk, which depend on the positions or weights w_i . Imagine multiplying all of the portfolio weights by the same constant, k , that is, consider the vector of portfolio weights $kw = (kw_1, kw_2, \dots, kw_N)'$ and the associated portfolio standard deviation $\sigma(kw)$ and value-at-risk $\text{VaR}(kw)$. A key property of the portfolio standard deviation is that scaling

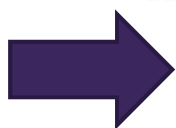
Extract from Neil D. Pearson, Risk budgeting: Portfolio Problem Solving with Value-at-Risk, Wiley 2002, pages 153-163

Risk Decomposition

all positions by the common factor k scales the standard deviation by the same factor, implying $\sigma(kw) = k\sigma(w)$. This is also true of the value-at-risk, because scaling every position by k clearly scales every profit or loss by k , and thus scales the value-at-risk by k .

In mathematical terminology, the result that $\text{VaR}(kw) = k\text{VaR}(w)$ for $k > 0$ means that function giving the value-at-risk is *homogenous of degree 1*, or *linear homogenous*. From a financial perspective, this property of value-at-risk is almost obvious: if one makes the same proportional change in all positions, the value-at-risk also changes proportionally.

Though very nearly obvious, this property has an important implication. If value-at-risk is linear homogenous, then Euler's law (see the notes to this chapter) implies that both the portfolio standard deviation and VaR can be decomposed as



$$\sigma(w) = \frac{\partial \sigma(w)}{\partial w_1} w_1 + \frac{\partial \sigma(w)}{\partial w_2} w_2 + \dots + \frac{\partial \sigma(w)}{\partial w_N} w_N \quad (10.1)$$

and



$$\text{VaR}(w) = \frac{\partial \text{VaR}(w)}{\partial w_1} w_1 + \frac{\partial \text{VaR}(w)}{\partial w_2} w_2 + \dots + \frac{\partial \text{VaR}(w)}{\partial w_N} w_N, \quad (10.2)$$

Risk Decomposition

respectively. The i th partial derivative, $\partial\sigma(w)/\partial w_i$ or $\partial\text{VaR}(w)/\partial w_i$, is interpreted as the effect on risk of increasing w_i by one unit; in particular, changing the i th weight by a small amount, from w_i to w_i^* , changes the risk by approximately $(\partial\sigma(w)/\partial w_i)(w_i^* - w_i)$, or $(\partial\text{VaR}(w)/\partial w_i)(w_i^* - w_i)$. The i th term, $(\partial\sigma(w)/\partial w_i)w_i$ or $(\partial\text{VaR}(w)/\partial w_i)w_i$, is called the *risk contribution* of the i th position and can be interpreted as measuring the effect of percentage changes in the portfolio weight w_i . For example, the change from w_i to w_i^* is a percentage change of $(w_i^* - w_i)/w_i$, and the change in portfolio standard deviation resulting from this change in the portfolio weight is

$$\frac{\partial\sigma(w)}{\partial w_i}(w_i^* - w_i) = \frac{\partial\sigma(w)}{\partial w_i}w_i \times \frac{(w_i^* - w_i)}{w_i},$$

the product of the risk contribution and the percentage change in the weight.

Risk Decomposition

A key feature of the risk contributions is that they sum to the portfolio risk, permitting the portfolio risk to be decomposed into the risk contributions of the N positions w_i . Similarly, we can define $((\partial\sigma(w)/\partial w_i)w_i)/\sigma(w)$ or $((\partial\text{VaR}(w)/\partial w_i)w_i)/\text{VaR}(w)$ to be the percentage contribution to portfolio risk of the i th position. It is straightforward to compute these risk contributions when risk is measured by standard deviation. Computing the derivative with respect to the i th portfolio weight,

$$\frac{\partial\sigma(w)}{\partial w_i} = \frac{\sum_{j=1}^N w_j \text{cov}(r_i, r_j)}{\sigma(w)}, \quad (10.3)$$

where the numerator $\sum_{j=1}^N w_j \text{cov}(r_i, r_j)$, is the covariance between the return r_i and the portfolio return $\sum_{j=1}^N w_j r_j$. Thus, the risk contribution of the i th position or asset

$$\frac{\partial\sigma(w)}{\partial w_i}w_i = \frac{\sum_{j=1}^N w_j \text{cov}(r_i, r_j)}{\sigma(w)}w_i$$

Risk Decomposition

is proportional to the covariance between the return on the i th position and the portfolio, and is zero either when that position is uncorrelated with the portfolio or when the weight $w_i = 0$. The percentage contribution to portfolio risk is

$$\begin{aligned}\frac{\partial \sigma(w)}{\partial w_i} \frac{w_i}{\sigma(w)} &= \frac{\sum_{j=1}^N w_j \text{cov}(r_i, r_j)}{\sigma^2(w)} w_i \\ &= \beta_i w_i,\end{aligned}$$

where β_i is the regression coefficient, or beta, of the i th market return on the return of the portfolio. By construction, the weighted sum of the betas is 100% or one, that is, $\sum_{i=1}^N \beta_i w_i = 1$. This is exactly analogous to the standard result in portfolio theory that the market beta of one is the weighted average of the betas of the stocks constituting the market.

Decomposing delta-normal value-at-risk is almost equally easy. Recognizing that delta-normal VaR is of the form

Risk Decomposition

$$\text{VaR}(w) = - \left[\sum_{i=1}^N w_i E[r_i] - k \sigma(w) \right],$$

the i th partial derivative and risk contribution of the i th asset are

$$\frac{\partial \text{VaR}(w)}{\partial w_i} = -E[r_i] + k \frac{\partial \sigma(w)}{\partial w_i}$$

and

$$\frac{\partial \text{VaR}(w)}{\partial w_i} w_i = -E[r_i] w_i + k \frac{\partial \sigma(w)}{\partial w_i} w_i,$$

respectively. The only difference (besides the constant k) between this and equation (10.3) is the expected return component $-E[r_i] w_i$, which appears because larger expected returns shift the distribution upward and reduce the probability of loss.

Risk Decomposition

The key insight from these results is that the risk contribution of a position crucially depends on the covariance of that position with the existing portfolio. This covariance is zero when the position is uncorrelated with the existing portfolio, in which case the risk contribution is zero. When the correlation is positive, the risk contribution is positive; when it is negative, the position serves as a hedge, and the risk contribution is negative.

In interpreting risk decomposition, it is crucial to keep in mind that it is a marginal analysis; a small change in the portfolio weight from w_i to w_i^* changes the risk by approximately $(\partial\sigma(w)/\partial w_i)(w_i^* - w_i)$, or $(\partial\text{VaR}(w)/\partial w_i)(w_i^* - w_i)$. Alternatively, if the risk decomposition indicates that the i th position accounts for one-half of the risk, increasing that position by a small percentage will increase risk as much as increasing all other positions by the same percentage. The marginal effects cannot be extrapolated to large changes, because the partial derivatives $\partial\sigma(w)/\partial w_i$ and $\partial\text{VaR}(w)/\partial w_i$ change as the position sizes change. In terms of correlations, a large change in market position changes the correlation between the portfolio and that market. For example, if the i th market is uncorrelated with the current portfolio, the risk contribution of a small change in the allocation to the i th market is zero. However, as the allocation to the i th market increases, that market constitutes a larger fraction of the portfolio and the portfolio is no longer uncorrelated with the i th market. Thus,

Risk Decomposition

the risk contribution of the i th market increases as the position in that market is increased.

RISK DECOMPOSITION FOR HISTORICAL AND MONTE CARLO SIMULATION

The expression $(\partial\text{VaR}(w)/\partial w_i)(w_i)$ for the risk contribution of the i th position suggests computing the risk contributions by perturbing the position weights w_i by small amounts and then recomputing the value-at-risk to obtain estimates of the partial derivatives $\partial\text{VaR}(w)/\partial w_i$ and then multiplying by the weights w_i . While it is a natural first thought, there are two severe disadvantages to this approach. First, if there are N positions, this requires $N + 1$ VaR calculations. Given that the computational burden is a significant drawback of the Monte Carlo approach, this is unappealing. Second, even if the computational burden does not rule out that approach, in practice the simulation error is too large to permit an accurate estimate of the partial derivatives and risk contributions. However, all is not lost, as a good approach is available.

Risk Decomposition

Trivially, the value-at-risk estimate can be characterized as the (negative of the) expected loss, conditional on the set of scenarios such that the (negative of the) expected loss is equal to the VaR. For example, if we compute a 95%-confidence value-at-risk using 100 simulation trials, the value-at-risk estimate is the loss given by the 95th worst scenario. Changing a portfolio weight w_i , then, has two effects. First, it affects the loss in the set of scenarios such that the (negative of the) expected loss is equal to the VaR. Letting $r_i^{1-\alpha}$ denote the return on the i th asset in the scenario used to compute a $(1 - \alpha)$ -confidence VaR, the loss in this scenario is $\sum_{i=1}^N w_i r_i^{1-\alpha}$ and the effect of changing w_i on the loss in this scenario is

$$\frac{\partial \left(\sum_{i=1}^N w_i r_i^{1-\alpha} \right)}{\partial w_i} = r_i^{1-\alpha}. \quad (10.4)$$

Second, the change in w_i might have an effect on the scenarios used to compute the value-at-risk. A bit of thought suggests that this latter effect will be small: the probability that a very small change in one of the weights w_i causes a different scenario to be the 95th (or 950th or 9500th) worst scenario will typically be zero and will almost certainly be small. In fact, this second effect disappears in the limit as the change in $w_i \rightarrow 0$, implying that

Risk Decomposition

the partial derivative can be computed from (10.4). The risk contribution of the i th position is then simply

$$\frac{\partial \text{VaR}(w)}{\partial w_i} w_i = r_i^{1-\alpha} w_i. \quad (10.5)$$

If more than one scenario is used to compute the value-at-risk, the term $r_i^{1-\alpha}$ on the right-hand side of (10.5) is replaced by the average return in the scenarios used to compute the value-at-risk.

Risk Decomposition

NOTES

The approach to risk decomposition in this book is that of Litterman (1996), which has become standard. Adding confusion, some (e.g., Mina and Xiao 2001) defy standard usage of the word *marginal* and switch the definitions of marginal and incremental risk. That is, in RiskMetrics, marginal VaR is the change in risk resulting from selling the entire position and incremental VaR measures the effect of a small change in a position (Mina and Xiao 2001: Sections 6.2–6.3).

Euler's law is obvious in the one-variable case, because only functions of the form $f(w) = bw$ are homogenous of degree 1, implying that, for such functions, $f(kw) = kbw$ and $\partial f(kw)/\partial k = bw = f(w)$.

To obtain Euler's law in the general case, one starts with the statement that the value-at-risk of a portfolio kw is $\text{VaR}(kw) = k\text{VaR}(w)$ and differentiates both sides of this equation with respect to k . This differentiation amounts to asking what the effect is of a proportionate increase in all positions.

Risk Decomposition

The right-hand side of the equation means that the portfolio value-at-risk is proportional to k and thus increases at the rate $\text{VaR}(w)$ as k changes. Thus, the derivative of the right-hand side is

$$\frac{\partial(k\text{VaR}(w))}{\partial k} = \text{VaR}(w). \quad (10.6)$$

To compute the derivative of the left-hand side, write it as $\text{VaR}(kw) = \text{VaR}(y_1, y_2, \dots, y_N) = \text{VaR}(y)$, where $y_1 = kw_1$, $y_2 = kw_2$, $\dots$, $y_N = kw_N$, and y is the vector $y = (y_1, y_2, \dots, y_N)'$. Then,

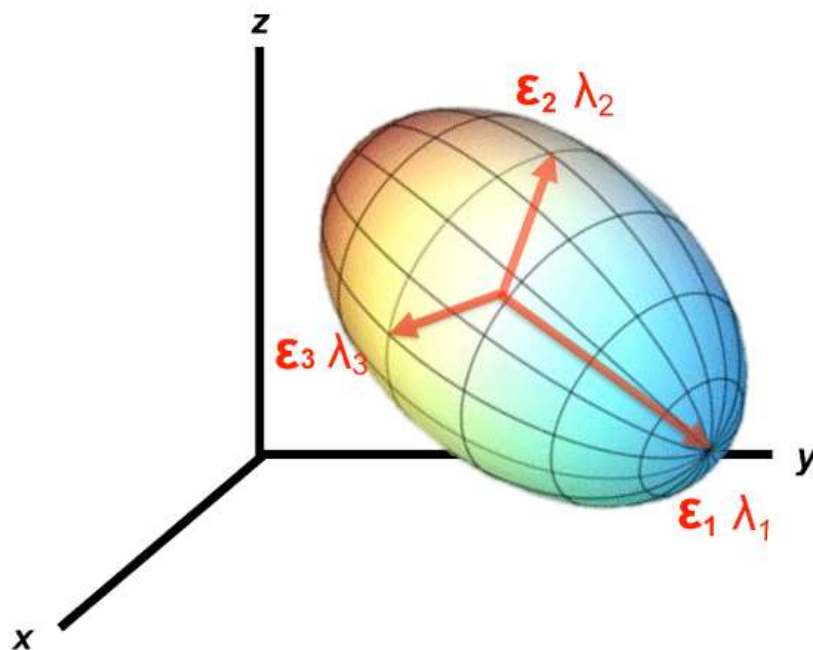
$$\frac{\partial \text{VaR}(y)}{\partial k} = \frac{\partial \text{VaR}(y)}{\partial y_1} \times \frac{\partial y_1}{\partial k} + \frac{\partial \text{VaR}(y)}{\partial y_2} \times \frac{\partial y_2}{\partial k} + \dots + \frac{\partial \text{VaR}(y)}{\partial y_N} \times \frac{\partial y_N}{\partial k}.$$

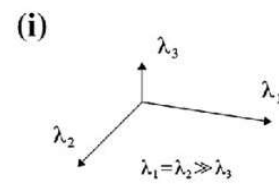
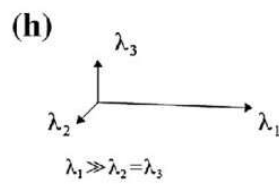
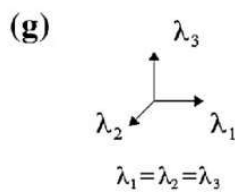
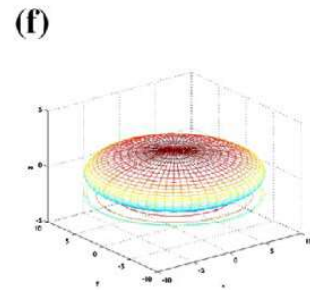
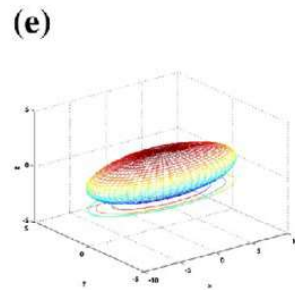
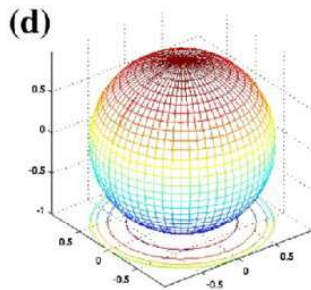
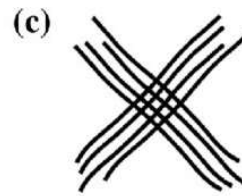
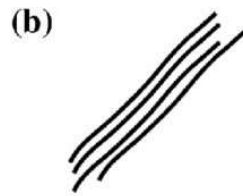
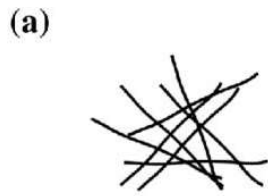
This says that the effect on the value-at-risk of increasing k is determined by the effects of y_i on VaR and the effects of k on y_i . Then, using the facts that $\partial y_1 / \partial k = w_1$, $\partial \text{VaR}(y) / \partial y_1 = \partial \text{VaR}(w) / \partial w_1$, and similarly for the other y_i s, one obtains

Risk Decomposition

$$\begin{aligned}\frac{\partial \text{VaR}(y)}{\partial k} &= \frac{\partial \text{VaR}(y)}{\partial y_1} w_1 + \frac{\partial \text{VaR}(y)}{\partial y_2} w_2 + \dots + \frac{\partial \text{VaR}(y)}{\partial y_N} w_N \\ &= \frac{\partial \text{VaR}(X)}{\partial w_1} w_1 + \frac{\partial \text{VaR}(w)}{\partial w_2} w_2 + \frac{\partial \text{VaR}(X)}{\partial w_N} w_N.\end{aligned}\quad (10.7)$$

Combining (10.6) and (10.7) yields equation (10.2) in the body of the chapter. A similar analysis will yield the decomposition of the standard deviation in equation (10.1)



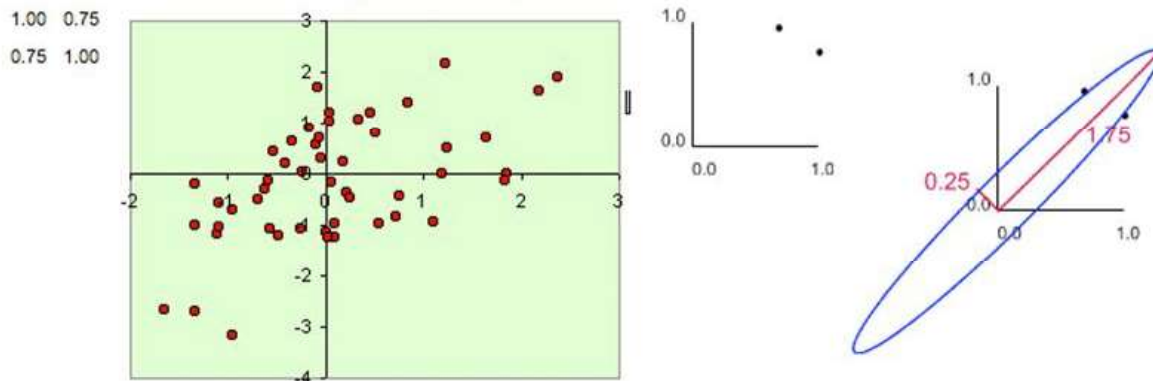


$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

Physical interpretation

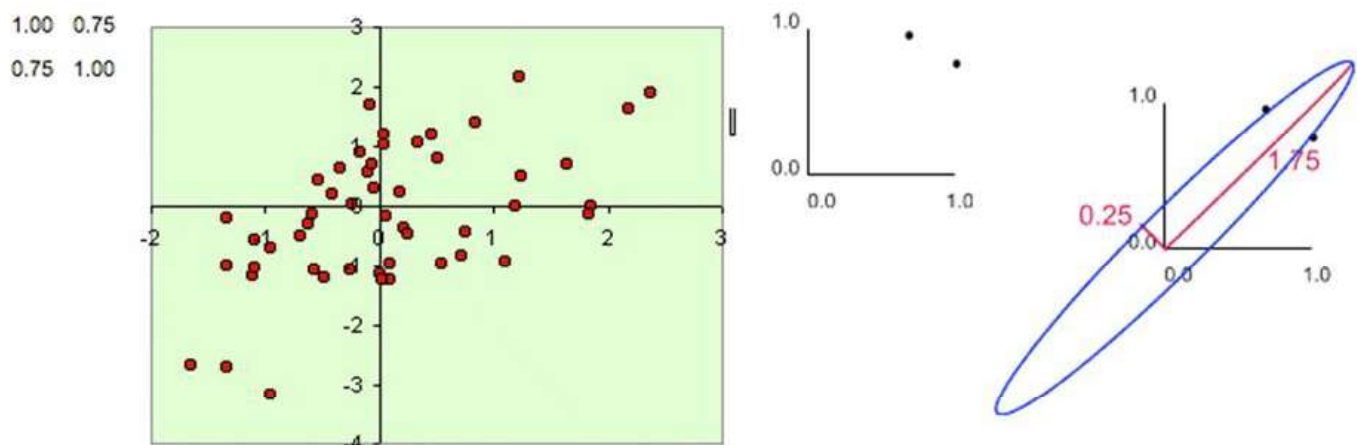
- Consider a covariance matrix, $\mathbf{A}$, i.e., $\mathbf{A} = 1/n \mathbf{S} \mathbf{S}^T$ for some $\mathbf{S}$

$$\mathbf{A} = \begin{bmatrix} 1 & .75 \\ .75 & 1 \end{bmatrix} \Rightarrow \lambda_1 = 1.75, \lambda_2 = 0.25$$



- Error ellipse with the major axis as the larger eigenvalue and the minor axis as the smaller eigenvalue

We can interpret this correlation as an ellipse whose major axis is one eigenvalue and the minor axis length is the other:



No correlation yields a circle, and perfect correlation yields a line.

Fixing a broken correlation matrix

Phil Joubert and Stephen Langdell

The correlation matrix is central to risk calculations, whether it is used as part of the definition of a copula or other dependency structure used to generate risk factors, or more directly to combine capital requirements using the “VCV” technique (e.g. in the Standard Formula method of Solvency 2). In this article we discuss a common problem: a matrix which superficially “looks like” a correlation matrix may not be one mathematically. First we review some theory of correlation matrices, before discussing what can go wrong. We then present some ways of constructing valid correlation matrices or “fixing” broken ones.

Correlation Matrix Theory

The correlation between two random variables describes the strength of any linear relationship between those variables on a scale which ranges from -1 to 1. We exclude the extremes of the scale as they would imply that the variables are deterministically related. For jointly normal random variables, this single number summarises all there is to know about the relationship between the two. For other distributions the situation can be more complicated, but we will not consider this. Correlations and correlation matrices are fundamentally a multivariate normal concept.

When more than two random variables (e.g. risk factors) are involved it is useful to draw up the correlations between each pair of values into a table. Appropriately set out this table looks like a matrix. We take the convention that the correlation of a variable with itself is 1 (the upper limit of the scale), and we note that the relationship is symmetric: the correlation between variables X and Y is the same as between Y and X .

From this we deduce that a correlation matrix is any matrix which:

1. Is symmetric
2. Has elements in $(-1,1)$ off the main diagonal
3. Has elements = 1 on the main diagonal.

But there is a fourth requirement. If we know the relationship between variables X and Y , and between Y and Z , then we should have some idea of the relationship between X and Z . All three have to be consistent.

For normal random variables we can try to force consistency by construction. If Z_1 and Z_2 are independent standard normal variables (i.e. their correlation and means are zero and their variances are one), then we can construct two dependent standard normal variables X_1 and X_2 by setting

Z and $X_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$, where ρ is the desired correlation. The first term builds the dependence, and the second is a “balancing item” to make sure the new variable (X_2) has variance one. We can continue by adding a third variable X_3 such that X_3 is a linear combination of Z_1 , Z_2 and Z_3 . This builds up a set of linear simultaneous equations which we can write in matrix form as

where L is a lower triangular matrix (i.e. it has zeroes above the main diagonal). Then we set the correlations of these variables to our desired correlation matrix C , giving us the equation

$$\text{Corr}[X] = E[X X^T] = E[(LZ)(LZ)^T] = LE[Z Z^T]L$$

The last identity is key here – a matrix C with properties 1, 2 and 3 above is only a correlation matrix if we can find a lower triangular matrix L such that $LL^T = C$. If it exists, this is called the Cholesky decomposition of C and is analogous to the square root of a real number. Matrices which have Cholesky decompositions also have several other characteristics which can be useful, and are called *positive definite* matrices. So we add a fourth characteristic:

4. Is a positive definite matrix.

Why a correlation matrix might be broken

Correlation matrices in some applications (e.g. portfolio risk) are calculated from historic data, but rarely in a consistent way. Data might be missing because a particular stock didn't trade on a given day, or a particular market was closed, or because the company didn't exist until five years ago. Correlations are calculated pairwise, and then put into the form of a symmetric matrix. There is no guarantee that this matrix satisfies requirement 4.

In insurance, lack of data means that a firm's correlation matrix is frequently set, at least partly, by "expert judgment". Managers might be asked to set correlations between risk factors as {high, medium, low, none} = {0.75, 0.5, 0.25, 0}. Even the technically inclined will struggle to keep the resulting matrix consistent (in the sense of requirement 4.) for a matrix of more than a couple of elements. For a typical case with dozens or even hundreds of risk factors this is almost impossible.

Whatever the origin of the problem, we are often presented with a broken correlation matrix and asked to fix it. Next we present a couple of repair techniques. We'll use a simple example to illustrate: the offending matrix is

1	0.95	0
	1	0.95
		1

Patch it up

A quick and dirty method of patching up a broken matrix exploits one of the characteristics of positive definite matrices: a positive-definite matrix always has positive eigenvalues (and vice-versa). Recall that a symmetric matrix C can be decomposed as

$$C = Q\Lambda Q^T$$

where Q is the matrix whose columns are the eigenvectors of C and Λ is a matrix with the corresponding eigenvalues on the diagonal and zero elsewhere. We don't need any more theory of eigenvalues than this, a decent statistical or numerical library will be able to calculate them for us¹.

We can attempt to fix the matrix by calculating the eigen-decomposition of the matrix, setting any negative eigenvalues to zero, and then reconstructing the resulting matrix. So we form $QA'Q^T$, where Λ' is the same as Λ , but with negative entries replaced by zero and scale the result to give a matrix C' with ones on the diagonal.

For our matrix the eigenvalues are $\{-0.3435, 1.0000, 2.3435\}$. Setting the first (negative) entry to zero and calculating C' as above results in

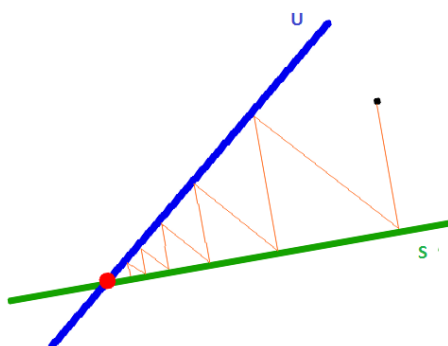
1.0000	0.73454	0.07908
	1.0000	0.73454
		1.0000

Find the closest correlation matrix

While the above method works, it is a bit arbitrary – we don't really understand its relationship to the original matrix. On the other hand, the closest correlation matrix method (Higham, 2002) explicitly minimises the distance between the original matrix and the “fixed” matrix. In this case we define the distance between two matrices as the square root of the sum of the squared differences of their elements:

$$\|X - Y\| = \sqrt{\sum \sum (x_{ij} - y_{ij})^2}$$

This is called the Frobenius distance. Computing the nearest correlation matrix to a given matrix is done numerically by iteratively and alternately projecting onto the spaces of positive definite and unit diagonal symmetric matrices, eventually converging to the closest matrix in the intersection of those spaces (see figure).



We find the matrix below²:

¹ The NAG library was used for this article

1.00000	0.75217	0.13151
	1.00000	0.75217
		1.00000

which looks similar to the matrix we found in the previous section. A quick computation of the Frobenius distance using the formula above would show that this matrix is indeed “closer” to the original.

Having an analytic framework to hang our method on means we can introduce more flexibility. Suppose the manager setting the correlations miscommunicated his intentions, and that the 0 in his matrix should have been interpreted as “I don’t care”. In this case we can add a weighting to the algorithm, so that the important correlations are disturbed as little as possible. Assigning appropriate weights we now get³:

1.00000	0.94044	0.82438
	1.00000	0.94044
		1.00000

Complete the matrix

The last scenario occurs often in practice, especially when dealing with large matrices. Matrices might be set in “blocks” by different business units, and need to be combined, or managers may be agnostic about the values of certain correlations, but certain about others. These are known as matrix completion problems.

Given the matrix

1.00	0.95	?
	1.00	0.95
		1.00

we can set the missing entry to the product of the known correlations, giving

1.00	0.95	0.9025
	1.00	0.95
		1.00

A proof that this does result in a positive definite matrix is given in Kahl and Günther (2005), which confirms the intuition that it should work. This method can be used to combine block matrices. It also suggests an alternative method of correlation matrix

² The NAG library algorithm (G02AAF) was used

³ The NAG library algorithm (G02AJF) was used

Numeric Software: Build or Buy?

As actuarial models get more sophisticated there is a growing requirement for more numerical and statistical methods, which need to be accurate, reliable and high-performance. Actuaries are often quite keen on building models in-house, but there are clear advantages to using proven algorithms from an established numerical library:

- Development time is reduced.
- Actuarial resource can be devoted to actuarial problems, rather than software problems.
- Code from a high-quality software house is exhaustively tested.
- The best algorithm for a given problem may not be the most obvious. For example, although the alternating projections algorithm for the nearest correlation matrix is guaranteed to converge, it is not the most efficient. The NAG Library codes use a Newton algorithm that is more complicated but converges much more quickly - especially for large matrices.
- Bugs can be subtle, especially since their effect can be masked by random noise in data.

construction, whereby business units are asked to provide pairwise correlations with one central random variable (say GDP growth or something similar), with all other correlations being inferred.

Conclusions

Correlation matrices are a key input to risk systems, and a major source of headaches. Technical actuaries and quants don't understand why business actuaries can't check that the matrices they provide are positive definite, and the business actuaries don't understand why the quants refuse to use a perfectly reasonable looking matrix. We've demonstrated a couple of ways out of this bind: either by finding a matrix which resembles the given matrix but which is positive definite, or by changing the method of specifying correlation matrices in the first place. These techniques are robustly implemented and we have successfully used them in large scale solutions for several insurers and other financial institutions.

Phil Joubert is an independent actuarial consultant specialising in implementation of risk models and blogs at www.not-normal-consulting.co.uk.

Stephen Langdell is a Senior Technical Consultant at the Numerical Algorithms Group

Note: A version of this article also appears in The Actuary magazine (September 2013).

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